

Greenhouse Climate Controller

Run the greenhouse controls for a plant that cannot complain, only wilt.

Win condition: Best plant-zone-control match supported by tour evidence.

THE SCIENCE YOU ARE USING

Controlled environments. A greenhouse lets growers set temperature, humidity, and light to match what a plant needs. Get it wrong and the plant wilts, scorches, or molds. Every setting is a tradeoff: more light can mean more heat and faster drying.

Evidence from the tour. On the greenhouse tour you gather clues about real plant needs and real zone settings. Back at the table you match plant cards to climate dials using that evidence, then defend the layout.

HOW TO BUILD AND RUN IT

- 01 Gather tour evidence.** During the greenhouse visit, note real temperature, humidity, and light clues for different zones.
- 02 Read the plant cards.** Each plant card lists needs and tolerances. Sort cards by what they require.
- 03 Set the climate dials.** Match each plant to a zone and set temperature, humidity, and light on the dial board.
- 04 Resolve tradeoffs.** When two plants want different things in one zone, choose the setting your evidence best supports.
- 05 Defend the layout.** Explain each placement using a specific clue from the tour.

Safety: Stay with tour leader. Do not touch restricted plants or greenhouse controls. Allergy: Plant and pollen sensitivity note.

MATERIALS

Plant profile cards	one set per team (4 teams, plus 1 spare)
Control dial boards	one per team (4, plus 1 spare)
Tour clue sheets	one per team (4, plus 2 spare)
Clipboards	one per team (4)
Dry erase markers	shared, 1 to 2 per team

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

Tour evidence we collected (temperature, humidity, light, per zone): _____

Our plant-to-zone placements and the dial settings we chose: _____

The hardest tradeoff we resolved, and the clue that decided it: _____

DEFEND YOUR DESIGN

What evidence made you change your design? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Correct setting choices	30
Use of tour evidence	25
Explanation	20
Layout efficiency	15
Teamwork	10
Total	100